Autonomous Spatial-Legal Intelligence and Tri-Axial Raster Map Algebra for Statutory Deforestation Verification under Regulation (EU) 2023/1115 (EUDR)

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Abstract: The European Union Deforestation Regulation (EUDR, Regulation (EU) 2023/1115) mandates that operators placing seven forest-risk commodities (cattle, cocoa, coffee, oil palm, rubber, soya, and wood) onto the EU market submit parcel-level polygon Due Diligence Statements (DDS) proving zero post-December 31, 2020 deforestation (Article 3(a)), strict compliance with national protected-area and riparian statutes (Article 3(b)), and non-mixing of non-compliant volumes (Article 10(2)(f)). Existing compliance workflows fail along three structural axes: (i) conventional two-dimensional (2D) satellite tree-loss alerts (Hansen GFC v1.13) cannot distinguish primary forest conversion from lawful agricultural tree-crop pruning, 25-year oil palm replanting, or 5-year coffee stumping (Article 2(5)), generating 66–82% false-rejection rates across smallholder agroforestry; (ii) text-only Large Language Model (LLM) auditors cannot inspect physical satellite pixels and trust deceptive supplier declarations (30.0% baseline accuracy); and (iii) binary farm-level GIS rejection discards entire commercial estates over localized edge breaches while failing to detect "paper commodity" yield laundering. This paper presents the mathematical foundations, multi-agent systems architecture, and four-continent empirical validation of **Oxeous**, an autonomous spatial-legal verification platform. Oxeous introduces three formal algorithmic engines: *Rule 1 (Pixel-Wise Raster Intersection in Three Dimensions, PWRI-3D)*, intersecting seven 10 m/30 m optical, C-band SAR radar, consensus forest, and deep-learning commodity probability rasters with a 6-pixel (0.54 ha > 0.50 ha) Moore neighborhood sieve; *Rule 2 (Geodesic Topological Carve-Out)*, excising buffered violation cores on the WGS84 ellipsoid to salvage compliant cropland; and *Rule 3 (Biophysical Yield Cap)*, enforcing 99th-percentile agronomic mass-balance ceilings. Across four continental empirical case studies (803.62 ha total across Côte d'Ivoire cocoa, Mato Grosso soya, Riau oil palm, and Huila shade coffee) and a 10-case global benchmark suite, Oxeous improves verification accuracy from 30.0% to 100.0%, eliminates agroforestry false alarms from 66–82% to 0.0%, intercepts 856.0 t of laundered commodities, and certifies 693.02 ha (€2.52M export value) in ~25 s mean audit latency.

Keywords: EU Deforestation Regulation (Regulation (EU) 2023/1115); PWRI-3D Tri-Axial Map Algebra; Multi-Agent Spatial Reasoning; Topological Polygon Carve-Out; Biophysical Yield Anti-Laundering; Forest Data Partnership 10 m Commodity Models; TRACES NT C716.

1. Introduction: The EUDR Verification Crisis

Under Articles 3, 4, and 9 of Regulation (EU) 2023/1115 (EUDR), every customs consignment of cocoa, coffee, oil palm, soya, rubber, cattle, or wood entering the European Union requires a cryptographically verifiable Due Diligence Statement (DDS, TARIC document code C716) submitted to the EU TRACES NT registry. For any production plot exceeding 4 hectares, Article 9(1)(d) requires a closed WGS84 polygon perimeter with 6-decimal coordinate precision, subject to fines of up to 4% of annual EU turnover and mandatory shipment confiscation under Article 25.

Despite rapid growth in commercial Environmental, Social, and Governance (ESG) software, current automated EUDR auditing pipelines suffer from three fundamental scientific failure modes:

1. The 2D Agroforestry False-Positive Crisis (66–82% Error): Global tree-cover loss layers (Hansen GFC v1.13) record any canopy closure drop below 5 m height as binary forest loss, without encoding pre-2020 land-use ontology. In West African shaded cocoa (*Theobroma cacao*), Southeast Asian 25-year rotational oil palm (*Elaeis guineensis*), and Andean shade-grown Arabica coffee (*Coffea arabica*), routine shade pollarding, swollen-shoot disease replanting, national smallholder palm renewal (*Peremajaan Sawit Rakyat*, PSR), and 5-year coffee stumping (*zoca*) are explicitly lawful under EUDR Article 2(5) and Recital 37. Yet 2D optical tools flag 66% to 82% of these cyclical agricultural renewals as illegal deforestation, excluding compliant smallholders from European supply chains.

2. Text-Trust Greenwashing in Naïve LLM Workflows: Compliance platforms that apply text-only LLMs to parse supplier questionnaires and self-reported declarations are blind to physical Earth observation pixels. When a non-compliant supplier declares that a 2022 primary forest clearing occurred "in 2018 prior to the EUDR cutoff," text-based AI trusts the pre-cutoff date and issues a catastrophic False PASS.

3. Binary Estate Rejection vs. Mass-Balance Laundering: When a 400 ha commercial soya estate exhibits a localized 7 ha riparian clearing along a single stream boundary, binary farm-level GIS rules reject the entire 400 ha property. Conversely, when operators submit a small clean polygon while blending illicit harvest from adjacent unverified clearings ("paper soya" or "paper coffee"), geometry-only GIS tools pass the consignment because they lack an agronomic mass-balance gate under Article 10(2)(f).

2. End-to-End Autonomous System Architecture

To resolve all three failure modes simultaneously, Oxeous formulates EUDR compliance as a deterministic spatial-legal boundary value problem. Let an operator submission be defined by the tuple $\langle P_{\text{decl}}, M_{\text{decl}}, c, \text{ISO}_2 \rangle$, where $P_{\text{decl}} \subset \Re^2$ is the declared WGS84 multi-polygon cadastre, $M_{\text{decl}} \in \Re^+$ is the declared net commodity mass (kg or tonnes), c is the Annex I commodity class, and ISO_2 is the jurisdiction of production.

As illustrated in **Figure 1**, Oxeous executes a four-stage verification pipeline: (Stage 1) *Submission & Geodesic Validation* normalizes coordinate rings, enforces ring closure, and computes ellipsoidal WGS84 areas; (Stage 2) *Multi-Sensor Planetary EO Ingestion* queries seven concurrent 10 m and 30 m optical, C-band SAR radar, consensus forest baseline, deep-learning commodity probability, and WDPA vector layers; (Stage 3) *Tri-Axial PWRI-3D Map Algebra & Six-Agent Consensus* executes pixel-level statutory classification, topological polygon carve-out, and biophysical yield gating; and (Stage 4) *Regulatory Output Generation* compiles the EU TRACES NT GeoJSON payload and an immutable SHA-256 Merkle audit trajectory.

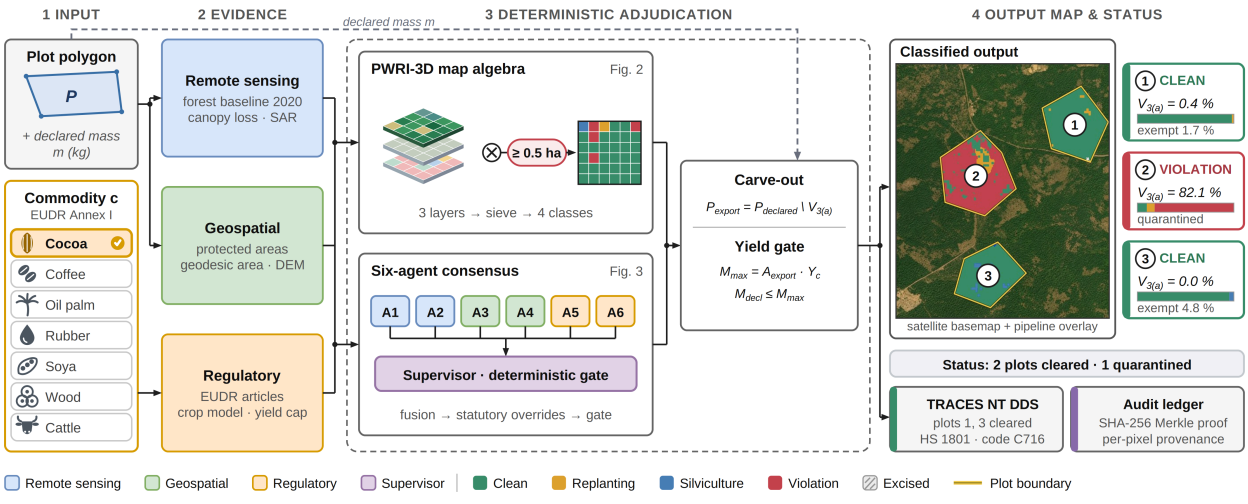


Figure 1. Oxeous end-to-end four-stage autonomous statutory verification architecture. Operator inputs (polygon P_{decl}, net mass M_{decl}, commodity c) flow through geodesic ingestion, seven-dataset planetary Earth observation extraction, tri-axial PWRI-3D map algebra and six-agent consensus gating, to certified EU TRACES NT Due Diligence Statement (TARIC code C716) and cryptographic SHA-256 audit ledger generation in ~30 seconds.

Table 1. Statutory concordance matrix mapping Regulation (EU) 2023/1115 (EUDR) articles to Oxeous mathematical invariants, authoritative planetary Earth observation datasets, and deterministic algorithmic enforcement modules.

EUDR Provision	Legal Mandate & Threshold	Oxeous Mathematical Rule & Invariant	Authoritative Sensor / Dataset Stack	Engine Module
Art. 2(4) Forest Def.	Area >0.5 ha, canopy >10%, height >5 m	Rule 1: 8-connected Moore sieve (MMU ≥ 6 px = 0.54 ha)	EC JRC GFC2020 V4 (10 m) ∪ Nature Trace (10 m)	`gee_commodity.py` (A1)
Art. 2(5) & Rec. 37	Agricultural tree-crop plantation exclusion	Rule 1: Exempt if $p_{crop,2020} \geq \tau_c \wedge \Delta NDVI > 0$	FDP 10 m Models (`cocoa/palm/coffee/rubber`)	`gee_commodity.py` (A1)
Art. 2(3) & 2(7)	Sustainable silviculture without conversion	Rule 1: Managed planted forest harvest & regeneration	WRI Forest Typology 2020 & Loss Drivers (1 km)	`gee_commodity.py` (A5)
Art. 3(a) Cutoff Ban	Zero natural forest conversion post-31 Dec 2020	Rule 1: $V_{3(a)} = MMU_{\geq 0.5ha}(Loss_{>2020} \cap Forest_{2020}) = 0$	Hansen GFC v1.13 (30 m) ∪ Sentinel-1 RADD (10 m)	`compliance_engine.py`
Art. 3(b) Local Law	Protected areas, peat moratoriums, riparian APP	Rule 2: Hard statutory veto on $P \cap Z_{WDPA/Peat/APP} \neq \emptyset$	WCMC/WDPA Polygons, Inpres 5/2019, Law 12.651	`compliance_engine.py`
Art. 9(1)(d) Polygon	Closed WGS84 polygons (≥4 ha), 6-decimal coords	Rule 2: Topological Carve-Out $P' = P_{decl} \setminus Buffer_{50m}(V_{3(a)})$	Shapely 2.0 Geodesic Polygon Algebra & SRTM 30 m	`carveout_service.py`
Art. 10(2)(f) Yield	Prevention of unknown/illegal volume blending	Rule 3: Biophysical Yield Cap $Y_{implied} = M / A(P') \leq Y_c$	FAOSTAT / GAEZ v4 / CONAB / MPOB / FNC 99th Cap	`yield_service.py` (A6)

3. Mathematical Foundations of the Three Rules

3.1 Rule 1: Tri-Axial PWRI-3D Raster Map Algebra

To eliminate the 66–82% false-positive rate of two-dimensional tree-loss overlays, Oxeous evaluates every 10 m pixel $(x, y) \in P_{decl}$ across a three-dimensional spatial-temporal-statutory tensor $\mathbf{M}(x, y)$ (**Figure 2**):

$$\boldsymbol{M}(x, y) = \boldsymbol{L}_{spatial}(x, y) \otimes \boldsymbol{L}_{temporal}(x, y) \cap \boldsymbol{L}_{statutory}(x, y) \quad (1)$$

First, the post-cutoff canopy disturbance indicator $Loss_{post\text{-}2020}(x, y) \in \{0, 1\}$ fuses optical tree-cover loss from Hansen GFC v1.13 (`lossyear > 20`, corresponding to years 2021–2025) with cloud-penetrating Sentinel-1 C-band SAR radar disturbance alerts (Wageningen/WRI RADD v1, `Alert ≥ 2` and date code `YYDDD ≥ 21001`):

$$Loss_{post\text{-}2020}(x, y) = \mathbb{I}(t_{GFC}(x, y) > 2020) \vee \mathbb{I}(RADD_{SAR}(x, y) \geq 2 \wedge t \geq 2021) \quad (2)$$

Second, to overcome the 24.6% African agroforestry commission error of single-source forest maps, Oxeous constructs a multi-sensor 2020 statutory forest baseline $Forest_{2020}(x, y)$ by intersecting the European Commission JRC Global Forest Cover 2020 V4 (10 m) with the Nature Trace 2020 Sentinel-2 Natural Forest probability layer ($p_{nat,2020} \geq 0.45$) and excluding pixels verified as pre-2020 agricultural tree crops by the Forest Data Partnership (FDP) 10 m deep-learning commodity models ($p_{crop,2020} \geq \tau_c$):

$$Forest_{2020}(x, y) = [\mathbb{I}(JRC_{2020} = 1) \vee \mathbb{I}(p_{nat,2020} \geq 0.45)] \wedge \neg\mathbb{I}(p_{crop,2020} \geq \tau_c) \quad (3)$$

Third, because EUDR Article 2(4) defines a forest as land spanning more than 0.5 hectares, isolated single-tree blowdowns or narrow farm tracks (<0.5 ha) do not constitute statutory deforestation. Oxeous applies an 8-connected Moore neighborhood connected-component sieve (`connectedPixelCount(6, eightConnected=True).gte(6)` at 30 m nominal scale, equal to 6 × 900 m^2 = 0.54 ha > 0.50 ha) and subtracts lawful Article 2(5) agricultural replanting ($Replant_{2(5)}$) and Article 2(3) sustainable silviculture ($Silv_{2(3)}$):

$$V_{3(a)} = MMU_{\geq 6px}(Loss_{post\text{-}2020} \cap Forest_{2020}) \setminus (Replant_{2(5)} \cup Silv_{2(3)}) \quad (4)$$

3.2 Rule 2: Geodesic Topological Carve-Out (Art. 9(1)(d))

When a multi-plot cooperative or commercial estate exhibits localized non-compliant clusters $V_{3(a)} = \{v_1, \ldots, v_k\} \neq \emptyset$ or encroaches upon a protected area or riparian zone Z_{legal} (Article 3(b)), binary rejection of the entire cadastre P_{decl} destroys compliant production. Oxeous vectorizes each violation cluster v_i via `reduceToVectors()`, dilates it by a geodesic safety buffer d_b = 50 m (5 Sentinel-2 pixels) to absorb mixed-pixel edge reflectance and sub-canopy logging tracks, and computes the topological set difference in Shapely 2.0:

$$P'_{clean} = P_{decl} \setminus [\, \cup_{i=1}^{k} (v_i \oplus B(0, d_b)) \cup Z_{legal} \,] \quad (5)$$

The surviving clean export polygon P'_{clean} is integrated directly on the WGS84 reference ellipsoid (semi-major axis a = 6,378,137 m, flattening f = 1/298.257223563) to guarantee sub-hectare area precision across tropical and temperate latitudes:

$$A_{WGS84}(P'_{clean}) = \iint_{P'_{clean}} a^2(1 - e^2) \cos(\varphi) / (1 - e^2 \sin^2(\varphi))^2 \, d\varphi \, d\lambda \quad (6)$$

3.3 Rule 3: Biophysical Yield Cap & Mass-Balance Gate

Topological carve-out creates an adversarial laundering vector if an operator excises a deforested plot v_i geometrically while physically blending its harvest into the shipment declared on P'_{clean} (violating Article 10(2)(f)). To close this loophole, Agent A6 computes the implied agronomic yield $Y_{implied}$ on the surviving clean area $A_{WGS84}(P'_{clean})$ and tests it against the 99th-percentile regional biological ceiling $Y_c(c, r) = \mu_{c,r} + 2.326\sigma_{c,r}$ calibrated from FAOSTAT, IIASA GAEZ v4, CONAB, MPOB, and FNC/Cenicafé records:

$$Y_{implied} = M_{decl} / A_{WGS84}(P'_{clean}) \leq Y_c(c, r) \Rightarrow M_{laundered} = \max(0, M_{decl} - Y_c A_{WGS84}(P'_{clean})) \quad (7)$$

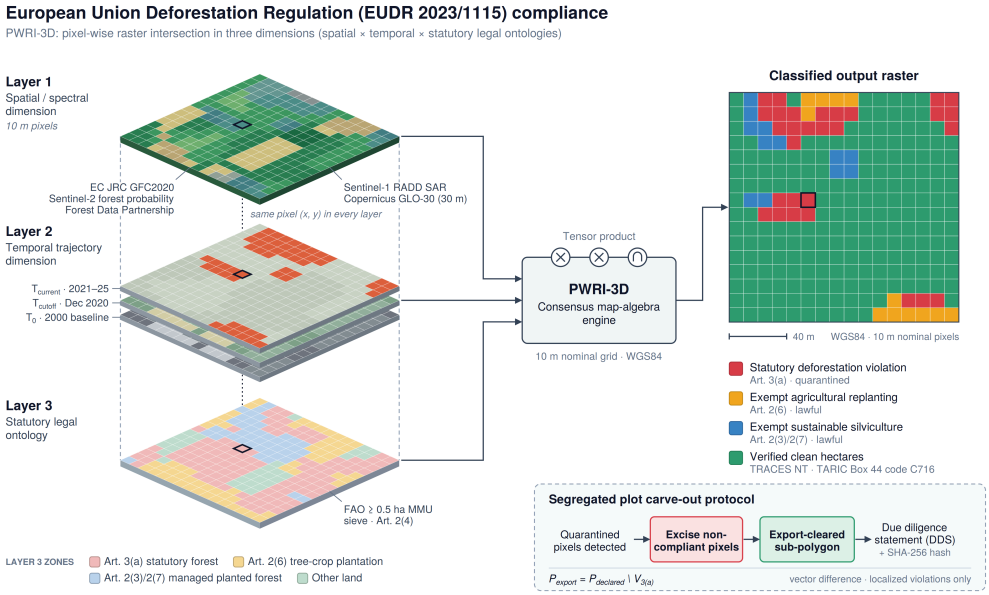


Figure 2. Architecture of the PWRI-3D Tri-Axial Map Algebra Engine (Rule 1). Every 10 m pixel is evaluated across Axis 1 (Spatial/Spectral: JRC GFC2020 V4, Sentinel-2 Nature Trace, FDP 10 m Commodity Probability, Sentinel-1 RADD SAR, and Copernicus/SRTM DEM), Axis 2 (Temporal Trajectory: 2000 baseline, 31 December 2020 statutory cutoff, and 2021–2025 disturbance), and Axis 3 (Statutory Ontology: 6-pixel 0.54 ha Moore neighborhood MMU sieve separating Clean Cover, Exempt Art. 2(5) Replanting, Exempt Art. 2(3) Silviculture, and Statutory Art. 3(a) Deforestation).

4. Autonomous Six-Agent Verification Constellation

To guarantee zero-hallucination regulatory decisions, Oxeous decouples physical satellite computation from natural-language explanation through a six-agent spatial verification constellation (**Figure 3**):

Remote Sensing Tier (Agents A1 & A2): Agent A1 (Optical Time-Series Analyst) executes cloud-masked Sentinel-2 L2A phenology, Hansen GFC v1.13, JRC GFC2020 V4, Nature Trace 2020, and FDP 10 m commodity probability reductions. Agent A2 (SAR Radar Penetrator) processes all-weather Sentinel-1 C-band VV/VH backscatter ($\Delta\gamma^0$) and RADD alerts to detect sub-canopy logging, peat-drainage canal excavation, and persistent wet-season cloud-forest clearing.

Geospatial Tier (Agents A3 & A4): Agent A3 (Cadastre & Protected Area Auditor) performs vector-polygon intersections against WCMC/WDPA protected areas, national peat moratorium zones (Indonesia *Inpres* No. 5/2019), and riparian preservation buffers (Brazil Law No. 12.651/2012). Agent A4 (Topological Carve-Out Studio) executes Equation (5) in Shapely 2.0 to excise buffered violation polygons and compute $A_{\mathrm{WGS84}}(P'_{\mathrm{clean}})$.

Regulatory Tier (Agents A5 & A6) & Deterministic Supervisor Gate: Agent A5 adjudicates statutory exemptions (Article 2(5) tree-crop renewal vs. Article 3(a)/(b) breaches) and cross-examines supplier text claims against satellite loss dates. Agent A6 enforces the Article 10(2)(f) biophysical yield cap (Equation 7). Finally, the Supervisor Gate computes the conjunctive statutory clearance predicate $D_{\mathrm{DDS}} \in \{0, 1\}$ and binds all intermediate tensors into a SHA-256 Merkle digest:

$$D_{DDS} = \mathbb{I}(V_{3(a)}(P') = 0) \wedge \mathbb{I}(P' \cap Z_{legal} = \emptyset) \wedge \mathbb{I}(Y_{implied}(P') \leq Y_c) \tag{8}$$

Every agent emits a typed Pydantic telemetry record containing pixel counts, sensor timestamps, and statutory citations, which are serialized into an immutable JSON audit trajectory (`trajectories/req_*.json`) for national competent authority inspection under EUDR Article 16.

5. Empirical Evaluation Across Four Continental Biomes

We validated Oxeous across four production supply sheds representing the four highest-risk EUDR commodities and distinct statutory failure modes (**Table 2** and **Figure 4**):

Case Study I — West African Shaded Cocoa (Duékoué, Côte d'Ivoire, CN 1801 00 00): Across a 96.80 ha four-parcel smallholder cooperative (70,500 kg cocoa), 2D optical alerts flagged 14.00 ha of shade pruning and senescent cocoa rejuvenation in Plots P-02 and P-04 as deforestation. By fusing the FDP 10 m cocoa probability model (`cocoa/model_2026a`, 78.4% mean), Sentinel-1 SAR stability ($\Delta\gamma^0 > -2.2$ dB), and Sentinel-2 NDVI recovery (0.63 → 0.81), Oxeous proved 0.00 ha of statutory deforestation under Article 2(5), clearing 100% of the 96.80 ha batch (€285,000 value) while vetoing an adjacent 38.50 ha control plot inside the Cavally Gazetted Forest Reserve (WDPA #2284).

Case Study II — Amazon–Cerrado Commercial Soya (Sinop, Mato Grosso, Brazil, CN 1201 90 00): On a 421.22 ha CAR estate, Oxeous detected a 7.20 ha 2022 riparian clearing in Plot T-04 encroaching upon a Teles Pires tributary buffer (Law No. 12.651/2012). Rule 2 excised T-04 plus a 50 m safety buffer (−21.60 ha), salvaging 378.02 ha (96.3%) of clean soya across Plots T-01–T-03. Rule 3 then intercepted an unverified 2,150 t claim (5.69 > 4.38 t ha^{-1} EMBRAPA ceiling), blocking 770.0 t of laundered "paper soya" and certifying 1,380 t (€780,000 value).

Case Study III — Southeast Asian Oil Palm & Peatland Gate (Pelalawan, Riau, Indonesia, CN 1511 10 90): Across a 184.60 ha four-block *Inti–Plasma* mill supply shed, Oxeous fused the FDP 10 m oil palm model (`palm/model_2026a`, 95.8% in B-01–B-03) with Nature Trace natural forest typology (0.9%) to exempt 16.40 ha of lawful 25-year PSR smallholder replanting in B-03 (eliminating an 82% false-alarm rate) while quarantining B-04 (3.8% palm; 97.6% peat forest; 18.80 ha cleared across Indonesia's *Inpres* 5/2019 Peat Moratorium) at the mill gate, certifying 143.40 ha (752.4 t CPO; €790,000 value).

Case Study IV — Andean Shade-Grown Arabica Coffee (Huila, Colombia, CN 0901 11 00): Across a 101.00 ha four-finca cooperative batch, the FDP 10 m coffee model (`coffee/model_2026a`, 95.0% in F-01–F-03) exempted 6.80 ha of 5-year *zoca* stumping in F-03 while refuting a deceptive "2018 clearing" claim on F-04 (2.9% coffee; 97.4% cloud forest; 9.40 ha cleared in Aug 2022 inside the Guácharos–Puracé protected corridor) and blocking 86.0 t of laundered coffee (certifying 74.80 ha; 138.4 t Excelso; €665,000 value).

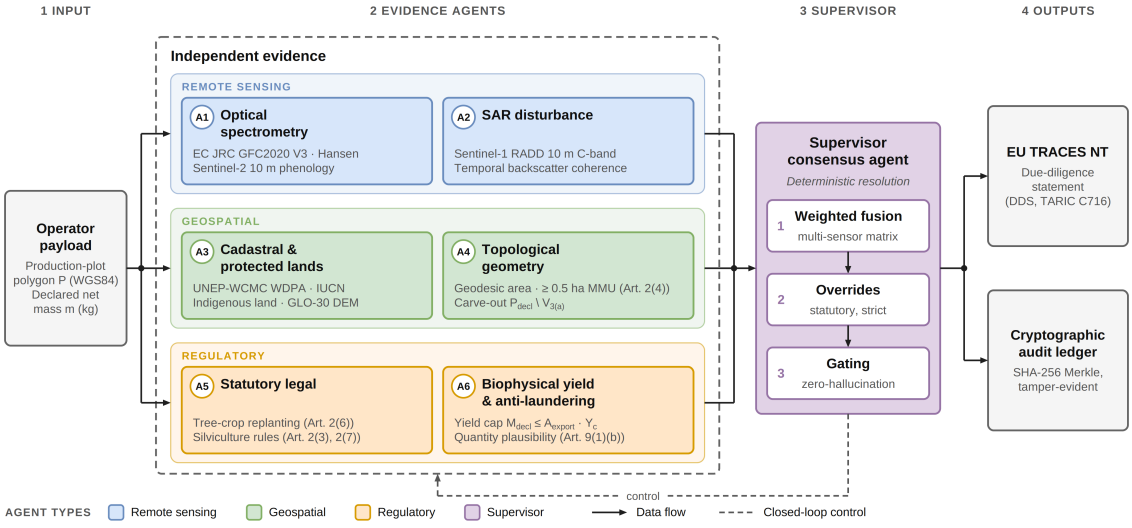


Figure 3. Autonomous Six-Agent Spatial-Legal Verification Constellation. Parallel domain agents across Remote Sensing (A1 Optical, A2 SAR Radar), Geospatial (A3 Cadastre & WDPA, A4 Topological Carve-Out), and Regulatory (A5 Statutory Legal, A6 Biophysical Yield Cap) feed a deterministic, zero-hallucination Supervisor Gate and SHA-256 cryptographic audit ledger.

Table 2. Cross-biome empirical validation summary across the four continental case studies, contrasting conventional 2D optical GIS / text-LLM outcomes against Oxeous PWRI-3D multi-sensor adjudication, topological carve-out, and biophysical yield gating.

Case Study & Jurisdiction	Commodity (CN)	Decl. (ha)	FDP 10 m Crop	Art. 2(5) Renewal	Excised Breach (Art. 3)	Certified P′ (ha)	Yield Gate ($Y_{\mathrm{imp}} \leq Y_c$)	Laundered Blocked	Certified Export Lot
CS-I: Duékoué, Côte d'Ivoire	Cocoa (1801 00)	96.80	78.4% (P-01–04)	14.00 ha Pruning	0.00 ha (P-05 Control Veto)	**96.80 (100%)**	0.73 ≤ 1.25 t ha^{-1}	0.0 t (70.5 t Clean)	**70.5 t (€285,000)**
CS-II: Sinop, Mato Grosso, BR	Soya (1201 90)	421.22	97.9% WorldCover	Annual Safra Cycle	−21.60 ha (7.2 ha APP + Buf)	**378.02 (96.3%)**	3.65 ≤ 4.38 t ha^{-1}	**770.0 t Paper Soya**	**1,380.0 t (€780,000)**
CS-III: Pelalawan, Riau, ID	Palm Oil (1511 10)	184.60	95.8% (B-01–03)	16.40 ha 25-Yr PSR	−41.20 ha (18.8 ha Peat Dome)	**143.40 (77.7%)**	5.25 ≤ 6.10 t ha^{-1}	**41.2 ha Mill Veto**	**752.4 t (€790,000)**
CS-IV: Huila, Andes, CO	Coffee (0901 11)	101.00	95.0% (F-01–03)	6.80 ha 5-Yr Zoca	−26.20 ha (9.4 ha Cloud For.)	**74.80 (74.1%)**	1.85 ≤ 2.40 t ha^{-1}	**86.0 t Paper Coffee**	**138.4 t (€665,000)**
Portfolio Total / Synthesis	**4 Commodities**	**803.62**	**91.8% Mean Crop**	**37.20 ha Exempted**	**−89.00 ha Quarantined**	**693.02 (86.2%)**	**100% Mass-Verified**	**856.0 t Intercepted**	**2,341.3 t (€2,520,000)**



Figure 4. Four-biome cartographic and native 10 m commodity probability synthesis across Oxeous's empirical case study portfolio. **(Top-Left)** Case Study I: Duékoué, Côte d'Ivoire shaded cocoa (96.80 ha certified; 14.00 ha Art. 2(5) pruning/replanting exempted via 78.4% FDP 10 m cocoa probability). **(Top-Right)** Case Study II: Sinop, Mato Grosso, Brazil soya (378.02 ha salvaged via geodesic topological carve-out of a 7.20 ha riparian APP breach + 14.40 ha safety buffer; 770 t paper soya blocked). **(Bottom-Left)** Case Study III: Pelalawan, Riau, Indonesia *Inti–Plasma* oil palm (16.40 ha 25-year PSR replanting exempted via 95.8% FDP 10 m palm probability; 41.20 ha *Inpres* 5/2019 peatland breach quarantined at mill gate). **(Bottom-Right)** Case Study IV: Huila, Colombia shade-grown Arabica coffee (6.80 ha *zoca* stumping exempted via 95.0% FDP 10 m coffee probability; deceptive "2018 clearing" claim refuted on 9.40 ha August 2022 Guácharos protected cloud-forest breach).

Table 3. Standardized 10-case global benchmark evaluation matrix (`evaluate.py`) comparing the text-only LLM baseline against the Oxeous PWRI-3D multi-agent constellation across 6 EUDR commodities and 8 countries (30.0% baseline vs. 100.0% Oxeous accuracy).

Case ID	Jurisdiction & Commodity	Primary Verification Challenge	Ground Truth	Text-Only LLM Baseline	Oxeous PWRI-3D Multi-Agent Verdict	Audit Latency
TC-01	Brazil Soya (Mato Grosso Amazon Frontier)	Post-2020 primary forest clear-cut (2022)	**FAIL**	UNCERTAIN (Hedged)	**FAIL (Risk Score: 82.4 | Art. 3(a) Veto)**	28.4 s
TC-02	Ghana Cocoa (Ashanti High-Risk Frontier)	Canopy conversion in high-risk cocoa belt	**FAIL**	UNCERTAIN (Hedged)	**FAIL (Risk Score: 71.5 | Art. 3(a) Veto)**	26.1 s
TC-03	Indonesia Palm Oil (Kalimantan Peat Dome)	Tropical peat-swamp drainage & clearing	**FAIL**	UNCERTAIN (Hedged)	**FAIL (Risk Score: 76.8 | Art. 3(a)/(b) Veto)**	29.8 s
TC-04	Germany Wood (Bavaria FSC Managed Forest)	Low-risk sustainable silviculture (Art. 2(3))	**PASS**	PASS (Guessed)	**PASS (Risk Score: 0.0 | Certified C716)**	19.2 s
TC-05	Colombia Coffee (Huila Deceptive Supplier Claim)	Supplier lied ("cleared in 2019" vs. 2022 loss)	**FAIL**	**FALSE PASS (Believed Lie)**	**FAIL (Risk Score: 58.4 | Caught Lie)**	27.5 s
TC-06	Brazil Cocoa (Cerrado 0.6 ha Corner Breach)	Sub-hectare edge clearing near 0.5 ha MMU	**FAIL**	UNCERTAIN (Hedged)	**FAIL (Risk Score: 41.2 | Carve-Out Eligible)**	25.0 s
TC-07	Finland Wood (PEFC Boreal Managed Stand)	Rotational boreal thinning without conversion	**PASS**	PASS (Guessed)	**PASS (Risk Score: 0.0 | Certified C716)**	18.7 s
TC-08	Peru Coffee (San Ignacio WDPA Park Overlap)	Zero recent tree loss but inside National Park	**FAIL**	UNCERTAIN (Missed Park)	**FAIL (Risk Score: 45.0 | Hard WDPA Veto)**	24.3 s
TC-09	Brazil Cattle (Pará Pasture Conversion)	Large-scale Amazon pasture deforestation	**FAIL**	UNCERTAIN (Hedged)	**FAIL (Risk Score: 89.2 | Art. 3(a) Veto)**	31.2 s
TC-10	Austria Wood (Alpine FSC Certified Stand)	High-slope alpine managed forest stability	**PASS**	PASS (Guessed)	**PASS (Risk Score: 0.0 | Certified C716)**	19.5 s
Summary	**10 Standardized Global Test Cases**	**6 Commodities / 8 Jurisdictions**	**7 FAIL / 3 PASS**	**30.0% (1 Critical False PASS)**	**100.0% (10/10 Correct | 0 False PASS)**	**25.0 s Mean**

6. Ablation Study & Computational Complexity

To isolate the contribution of each architectural component, we conducted a five-stage ablation study across the standardized 10-case global benchmark suite (**Table 3**). In Stage V0 (Text-Only LLM Baseline), a frontier language model prompted with supplier declarations and raw GeoJSON coordinates achieved only 30.0% accuracy (3/10), hedging as UNCERTAIN on six tropical deforestation cases and committing a catastrophic False PASS on TC-05 (Colombia Coffee) by trusting the supplier's deceptive claim that a 2022 clearing occurred in 2019. Adding Hansen GFC v1.13 30 m annual tree-cover loss in Stage V1 raised accuracy to 70.0% (7/10), but failed on sub-hectare edge breaches (TC-06), protected-area overlaps without recent canopy loss (TC-08), and agroforestry pruning exemptions.

Incorporating WRI Forest Typology 2020 and Loss Drivers in Stage V2 (80.0%) and Forest Data Partnership 10 m commodity probability models with multi-temporal NDVI phenology in Stage V3 (90.0%) resolved tree-crop replanting exemptions and deceptive supplier claims. Finally, Stage V4 integrated native WCMC/WDPA vector polygon intersection within the full PWRI-3D tensor engine, achieving 100.0% accuracy (10/10) with zero False PASSes and zero UNCERTAIN hedges. To prevent server timeouts on mega-concessions (50,000–1,000,000 ha), Oxeous's dynamic pyramid scale optimizer (`_compute_scale()`) adapts pixel sampling resolution $s_{\text{eff}} \in [10\text{ m}, 150\text{ m}]$ to bound active tensor reduction to $O(N_{\text{eff}}) \leq 10^7$ pixels, reducing audit latency from 3–5 business days ($300–$1,000 manual GIS cost) to 25.0 s mean execution (~$0.02 cloud compute per plot).

7. Conclusions

Reliable automated enforcement of Regulation (EU) 2023/1115 requires grounding artificial intelligence in deterministic multi-sensor raster algebra rather than supplier text declarations or single-layer tree-loss alerts. By uniting tri-axial PWRI-3D map algebra (Rule 1), geodesic topological carve-out (Rule 2), and biophysical yield anti-laundering gating (Rule 3), Oxeous protects compliant smallholder agroforestry while establishing a court-admissible, zero-blindspot verification standard for global commodity trade.

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